Fiscal Year:	FY 2010	**Task Last Updated:**	FY 09/09/2010
PI Name:	Mitchell, Cary Ph.D.		
Project Title:	(NSCORT) for Advanced Life Support at Purdue/Howard/AAMU - Crop and Bio Mass Production		
Division Name:	Human Research, Space Biology		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	West Lafayette	**State:**	IN
Zip Code:	47907-2010	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSCORT
Start Date:	12/01/2002	**End Date:**	11/30/2009
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	2
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA HQ
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 11/30/2009 per PI (1/09) NOTE: Received NCE to 11/30/2008 per PI (11/07)		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Beyl, Caula (Alabama A & M) Mitchell, Cary (Purdue University) Brown, Paul (Purdue University)		
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Three NSCORT (NASA Specialized Center of Research and Training) projects address the feasibility of producing edible biomass in a space regenerative life-support system. One project addresses a novel lighting approach to significantly reduce the ESM (Equivalent System Mass) of growing food crops in controlled environments; the second project tests the potential to use fungi to degrade non-edible crop residues while generating edible mushrooms; and a third project addresses the feasibility of feeding crop or bioreactor residue to fish that can be eaten. The fungi and fish projects were completed in 2006. Through 2009, the objective of the ALS NSCORT biomass production group is to minimize equivalent system mass (ESM) for food production in space habitats by conserving mass, volume, energy, and labor for component sub-systems of crop production through innovation leading to new technology and efficient practices. Developing revolutionary plant-lighting systems in space enabled by the many advantages of using light-emitting diodes (LEDs) will reduce all components of ESM. Developing effective canopy management and lighting architectures for vining species like sweetpotato will reduce the footprint for growing this valuable carbohydrate/anti-oxidant crop in space, and optimizing daylength and temperature regimes for strawberry will save energy and reduce labor for growing this dietary-enhancement fruit crop in space. Developing methods for rapid optimization of the plant-growth environment based upon real-time monitoring of photosynthetic gas exchange by crop canopies will save energy-intensive resources such as lighting and enhance the efficiency of food production in space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	1. The LED plant-lighting technologies being developed for significant ESM reduction for growing food crops in space during the NSCORT also can be used to make controlled-environment agriculture (CEA) economically viable on Earth. The technology can be used either for sole-source crop lighting or to supplement solar lighting in a greenhouse environment. The “smart-lighting” aspect of LED technology developed during the NSCORT can be applied just as well on the ground as in space to specifically target photosynthetic tissues and not waste energy lighting empty spaces. Developing effective shoot-management practices for highly branching, vining species like sweetpotato not only minimizes growth area occupied in a space habitat and ESM for autonomous life support, but will do the same in CEA on Earth, where lighted growth area also is very costly. The finding that berry yield of ‘Seascape’ strawberry is not different between 10 and 20 hours of daylength per 24-hour cycle allows flexibility in choosing photoperiods for a space-deployed salad machine growing a mix of crops with different light requirements, and also helps define lighting conditions that permit substantial savings of electrical energy for lighting without compromising plant yield. 2. Developing methods for maximum crop yield at minimum cost in controlled-environment agriculture (CEA) 3. Developing genetic engineering of CEA crops for superior performance
Task Progress:	Strawberry: Day-neutral strawberry plants (‘Seascape’) were grown long-term under 10-h photoperiods in three reach-in chambers set to day/night temperatures (°C) of 16°/8°, 18°/10°, or 20°/12°. Plants were grown for 282 days with fruit collection beginning at 65 days after planting. Fruit were collected, weighed and counted, and pH, Brix, and titratable acidity were measured. Overall, the warmest two temperature regimes produced the highest number of fruit, but the middle temperature regime, 18° day/10° night, produced the highest fruit fresh weight. Yields oscillated over time, with plants grown under warmer conditions fruiting earlier than those under cooler conditions. Brix averaged 10 for the first half of the harvest period but dropped to 8.4 during the latter half, indicating fruit may become less sweet as plants age. Titratable acidity averaged 0.6 but increased slightly during the middle of the harvest period. The highest yielding group (18°/10°) generally had the lowest Brix and highest tritatable acidity, indicating that fruit became less sweet and more tart over time. Fruit from the coolest treatment generally had higher Brix and lower titratable acidity and were slightly higher in pH than fruit from the other treatments. Sweetpotato: Sweetpotato vines were grown with different fertilizer concentrations. Vines were planted two-per-pot in ProfileTM calcined clay and maintained as single vines trained around cylindrical wire frames to reduce the area occupied. Plants were fertigated with Scotts Miracle Gro® Excel® Cal-Mag, a 15-5-15 commercial fertilizer formulation, at 1000 mg•L-1. This standard fertilizer solution contains (in mg•L-1) 200 N, 29 P, 167 K, 67 Ca, 30 Mg, and micronutrients, and is pH adjusted to 6.0 - 6.3 and alkalinity to 100 mg/L CaCO3. Plants were treated with 100% fertigation solution and dilutions of 75%, 50%, 25%, or 0% in one experiment, and a second experiment examined 10%, 20%, 30%, 40%, or 50%. Dilutions were made with tap water acidified to pH 6.2. In terms of root yield, 75% and 50% concentrations exceeded 100% in the first experiment, but yields declined sharply at 25% concentration of the fertigation solution. Stem weight declined gradually with fertilizer concentration from 100% down to 25% and then fell sharply with 0%, so root:shoot ratio was higher at all fertilizer concentrations compared to 100%. In a second experiment, root yields decreased gradually between 50% and 20%, with a large decline when 10% was used. Stem weight also declined as concentration decreased between 50% and 10%, so shoot-to-root ratios were generally consistent. Concentrations in the range of 50-75%, therefore, produce the best root yield while controlling stem growth, whereas concentrations below 50% limit root productivity. Tomato: Two cherry-tomato cultivars (‘Small Fry’, ‘Husky Cherry Red’) were grown hydroponically comparing intracanopy (IC) or overhead (OH) LED lighting. Results demonstrated much more edible and inedible biomass yield by the ‘Husky Cherry’ cultivar (209% more edible and 979% more nonedible dry weight compared to ‘Small Fry’). Plants grown with IC lighting produced 229% (dry weight) more biomass than those grown under OH lamps, and the more-productive ‘Husky Cherry’ also produced 126% (dry weight) more fruit with IC lighting than with OH. ‘Small Fry’ was significantly more sensitive to high-temperature stress and developed leaf intumescence (oedema), likely due to lack of UV light coming from the LED arrays. This comparative finding indicates that crops grown under narrow-wavelength solid-state lighting may have to be tested cultivar-by-cultivar. LED Lighting and Canopy Gas Exchange: A custom-designed crop gas-exchange cuvette was fabricated for use with intracanopy LED lighting. The cuvette will support 0.25 m2 of hydroponically grown planophile crops. There are 16 vertical LED strips, each with red (640nm) and blue (440nm) LEDs that are controlled remotely. Light engines are energized from the bottom up to keep pace with increases in crop height, and light output from red and blue LEDs is adjusted independently for variable red : blue ratios. The LED strips are reconfigurable into several different lighting architectures. Choice of several ceiling-slot configurations and plant positions in the lid of the hydroponic tub allow various arrangements of the lights and plants. Cuvette design allows for easy reconfiguration from one experiment to another, and additional ceiling and lid plates with novel configurations will be machined as needed. Graduate students in mechanical engineering technology and electrical engineering technology assembled and integrated structural

	components, sensors, and control systems, which include a USB video camera for internal monitoring of crop-stand growth, Vaisala temperature and humidity probes, differential and absolute IRGAs and mass flow controllers allowing rapid equilibration to CO2 set points, and real-time measurement of gas exchange throughout crop production. Trouble shooting and debugging of the gas exchange system will allow gas-exchange measurement of intracanopy, LED-lit candidate crop species to be performed during subsequent projects.
Bibliography Type:	Description: (Last Updated: 04/16/2020)
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